



FEU INSTITUTE OF TECHNOLOGY

COURSE CODE	COURSE TITLE	UNITS / TYPE
CS0025	SOFTWARE ENGINEERING 1	3 UNITS/LEC
PREREQUISITE/S	INFORMATION MANAGEMENT, ALGORITHM	
CO-REQUISITE/S	NONE	
COURSE DESCRIPTION	The major focus of this course is team-based collaboration of 3 to 4 students in the development of a CS project. The course covers fundamental concepts of Information Systems Analysis and Design, different Types of Information System, the System Development Life Cycle, the Scrum Methodology, Team Roles and Responsibilities, the Analysis and Design Process and the Methodologies, Models, Tools and Techniques supportive of the stages of typical System Development Life Cycle. The basics of Software Project Management will also be considered and emphasis on the development of deliverables throughout the Design Process leading to the development of the project.	

INSTITUTION VISION STATEMENT

FEU Institute of Technology aims to be the global standard of innovation in education, research, and nation-building by creating the next generation of changemakers that will shape the future of the country and the world.

INSTITUTION MISSION STATEMENT

FEU Institute of Technology is dedicated to provide quality, relevant, innovative and industry-based education producing competent and principled professionals with greater sense of responsibility, social awareness and high competitiveness contributing significantly to the betterment of the society.

EDUCATIONAL PHILOSOPHY

FEU Tech's educational philosophy is rooted in transformative learning, emphasizing innovation, relevance, and social responsibility. We aim to cultivate an ecosystem of learning that shapes the next generation of leaders and innovators, driving positive change both locally and globally.

DEPARTMENT VISION STATEMENT

The Computer Science department provides specialized programs in Computer Science whose graduates are internationally recognized as innovative and specialized computing professionals.

DEPARTMENT MISSION STATEMENT

The Computer Science department is committed to provide the students and faculty with an open environment that imbues professional and personal growth through research and development.

PROGRAM EDUCATIONAL OUTCOMES

The graduates of the Bachelor of Science in Computer Science program are:

1. engaged in further professional development and have interest in or aptitude for advanced studies or trainings in computing.
2. entrepreneurs or are employed in computing industries, organizing and managing team-based projects leading to successful and sustainable computing systems solutions.

3. responsible computing professionals actively participating in community groups that make a significant impact in addressing current and future societal challenges.

PROGRAM OUTCOMES

- a. Apply knowledge of computing appropriate to the discipline.
- b. Analyze a complex problem and identify and define the computing requirements appropriate to its solution.
- c. Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.
- d. Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal
- e. Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.
- f. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions
- g. Analyze the local and global impact of computing on individuals, organizations, and society.
- h. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.
- i. Apply knowledge through the use of current techniques and tools necessary for the IT profession.
- j. Apply mathematical foundations, algorithmic, principles, and computer science theory in the modeling and design of computer based systems in a way that demonstrate comprehension of the tradeoffs involved in design choices and applicable standards.
- k. Apply design and development principles in the construction of software systems of varying complexity in accordance with applicable standards.
- l. Apply mastery on their field of specialization

GRADUATE ATTRIBUTES

These are the skills and qualities our computing graduates will have when they finish the program.

- | | |
|---|--|
| 1. Academic Education | 6. Individual and Team Work |
| 2. Knowledge for Solving Computing Problems | 7. Communication |
| 3. Problem Analysis | 8. Computing Professionalism and Society |
| 4. Design/ Development of Solutions | 9. Ethics |
| 5. Modern Tool Usage | 10. Life-long Learning |

SMART-AIMED FRAMEWORK ALIGNMENT

This syllabus follows the SMART-AIMED Framework: **S**tudent-Centric, **M**entor-Guided, **AI**-Powered, **R**einforced, and **T**ransformative learning through **A**daptive, **I**ntegrated, **M**entor-Led, **E**vidence-Driven, and **D**emonstration of Transfer learning experiences. The framework supports Outcome-Based Education (OBE), purposeful AI integration, and authentic assessment.

AI is used as a learning partner to deepen learning and support mastery, feedback, and problem-solving in accordance with the AI Use and Academic Integrity Policy. Students are expected to use AI responsibly, ethically, and transparently and comply with the permitted, restricted, and prohibited uses specified for course activities and assessments.

COURSE LEARNING OUTCOMES AND RELATIONSHIP TO PROGRAM OUTCOMES

COURSE LEARNING OUTCOMES (CLO)	PROGRAM OUTCOMES											
	a	b	c	d	e	f	g	h	i	j	k	l
1. Analyze user and system requirements and design appropriate software solutions using software engineering principles, SDLC, UML, and system modeling techniques.			D									
2. Collaborate effectively as a member or leader of a software development team by applying Agile project management practices, defined team roles, and professional communication during project development.				E								

		Innovation Center Orientation - Overview of FTIC's innovation programs Academic Integrity - Pledge Principles of academic honesty and integrity AI Responsible Use Orientation - Discussion on AI Policy and Responsible Use Self-Introduction Activity - Student personal and academic background - Initial interaction and engagement with classmates	expectations and communication channels. Academic Integrity Pledge: - Discuss the pledge of honor and confirm commitment to uphold academic honesty and integrity in all course activities. Self-Introduction Activity: - Students introduce themselves by sharing relevant information (e.g., background, interests, expectations) and engage with their classmates.		
Week 1-2 / 6 hours	MODULE 1 – INTRODUCTION AND FUNDAMENTALS				
	- To explain the fundamental concept, principles and process framework of Software Engineering - To differentiate various types of Information System and their business applications. - To apply the phases of the SDLC in solving software development problems	Software Engineering Fundamentals - Software Categories - Legacy and Infant Software - Software Process Framework - Software Engineering Principles Types of Information Systems - IS Activities - Resourch Approach - Information Management SDLC Phases - SDLC vs Software Development Life Cycle	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> Faculty-Led Discussion: - Explain software engineering fundamentals, software categories, software process framework, and SDLC through an interactive lecture with guided questioning. Canvas Activities: - Complete a pre-class diagnostic quiz and submit a reflection comparing software engineering and traditional programming. Asynchronous Learning through Paraverse:	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> <u>Mastery Check (AI-Assisted):</u> Students demonstrate mastery of the lesson by completing an AI-supported assessment that applies concepts, verifies understanding, and documents responsible AI use. <u>Summative 1:</u> Guided Diagnostic on Software Engineering Fundamentals and SDLC (individual, supervised) <u>(AI-Prohibited)</u>	1

			• M-Flix video, complete Flashcards, and explore the Network Map on Software Engineering concepts Diagnostic Mapping: • Readiness quiz on prerequisite knowledge (Information Management and Algorithms). Adaptive Practice: • Solve SDLC case scenarios and receive instructor feedback before class discussion. AI-Assisted Activity: • Use ChatGPT to generate an SDLC example for a simple system, then identify inaccuracies and revise the lifecycle based on course concepts before submission. <i>(Students must verify AI output.)</i>	<u>Technical Assessment 1:</u> Feasibility Study for Target Thesis Title Proposals <u>(AI-Assisted)</u> <u>Technical Assessment 2:</u> RRL and RRS <u>(AI-Assisted)</u> Evidence of Learning: • Feasibility Study • RRL and RRS • Quiz (Summative)	
Week 3 / 4 hours	MODULE 2 – SOFTWARE PROJECT				
	• To analyze software project management processes using appropriate planning tools. • To evaluate system analysis techniques in identify software project requirements.	Software Project ○ What is a Project ○ Project Management ○ Gantt Chart and PERT Chart ○ Four Ps Approach ○ Project Management Approach and Phases ○ Project Documentation and Characteristics ○ Umbrella Activities System Analysis Overview ○ Analysis and Synthesis	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> Faculty-Led Discussion: • Discuss project management, Gantt and PERT charts, Four Ps, and software project documentation. Canvas Activities: • Submit a Gantt chart exercise and participate in a discussion forum on project management challenges. Asynchronous Learning through Paraverse: • Complete M-Flix videos on project	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> <u>Technical Assessment 3:</u> Formulation of First Title Proposal <u>(AI-Assisted)</u> Evidence of Learning: The group submits: • Title Proposal (First)	1, 2

		○ System Analysis Phase and Procedure ○ Computer Science Research Trends	management and review Flashcards. Diagnostic Mapping: ● Short readiness exercise interpreting a project scenario Adaptive Practice: ● Prepare a preliminary project schedule and receive instructor feedback. AI-Assisted Activity: ● Use AI to generate a project schedule, then revise the schedule by identifying unrealistic timelines, dependencies, and resource allocations.		
Week 4 / 4 hours	MODULE 3 – AGILE METHODOLOGIES				
	● Compare Agile software development methodologies and recommend an appropriate development approach for a given project. ● To apply Agile principles and Scrum practices in software project planning.	Agile Methodologies and Scrum ○ Agile ○ Agile Development ○ Agile vs Waterfall Crystal Methodology, DSDM, LSD, FDD and Extreme Programming ○ Crystal Methodology ○ DSDM Techniques ○ FDD, LSD and Extreme Programming	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> Faculty-Led Discussion: ● Explain Agile principles, Scrum framework, and Agile methodologies. Canvas Activities: ● Participate in a Scrum discussion forum and submit sprint planning worksheets. Asynchronous Learning through Paraverse: ● Review Agile M-Flix videos and Flashcards. Diagnostic Mapping: ● Compare Waterfall and Agile through a readiness activity. Adaptive Practice: ● Conduct a Scrum simulation with assigned team roles.	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> Summative 2: Agile Methodologies and Design Diagrams and System Requirements <u>(AI-Prohibited)</u> Technical Assessment 4: Comparative Analysis of Agile Methodologies (Agile vs. Waterfall or Scrum vs. XP). <u>(AI – Assisted)</u> Technical Assessment 5: Formulation of the Second Title Proposal <u>(AI-Assisted)</u> Evidence of Learning: Individual submission: ● Quiz (Summative) ● Comparative Analysis of Agile Methodologies	2

			AI-Assisted Activity: Prompt AI to generate a sprint backlog, then critique, prioritize, and improve the backlog based on Scrum principles.	Group submit: Second Title Proposal	
Week 5 / 4 hours	MODULE 4 – DESIGN DIAGRAMS AND SYSTEM REQUIREMENTS				
	- To develop software models and design diagrams to represent software requirements and business processes - To evaluate software requirements using appropriate requirements analysis techniques	Modeling Tools and Techniques - System Models - System Modeling Types - System Architecture - Business Model - Enterprise Modeling - Data Modeling - Data Flow Diagram System Requirements Checklist - System Requirements - Typical Software Checklist - Requirements Analysis - Fact Finding Techniques	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> Faculty-Led Discussion: - Demonstrate system models, DFDs, business process models, and requirements analysis. Canvas Activities: - Submit DFD diagrams and participate in peer review. Asynchronous Learning through Paraverse: - Complete M-Flix videos and Flashcards on DFDs and requirements engineering. Diagnostic Mapping: - Identify errors in a sample DFD. Adaptive Practice: - Create Level 0 and Level 1 DFDs for a case study. AI-Assisted Activity: - Use AI to generate a DFD or requirements list, then identify missing processes, incorrect data flows, and incomplete requirements before revising the model.	<i>The following are common across all modules. Faculty may add other TLAs appropriate to the course</i> Technical Assessment 6: Formulation of the Third Title Proposal (AI-Assisted) Technical Assessment 7: Formulation of the Fourth and Fifth Title Proposal (Optional) (AI-Assisted) Evidence of Learning: The group submits: - Third Title Proposal - Fourth Title Proposal and Fifth Title Proposal (Optional)	1, 3

Week 6 / 2 hours	MIDTERM EXAMINATION (AI-PROHIBITED)				
Week 7 / 4 hours	MODULE 5 – OBJECT – ORIENTED ANALYSIS AND DESIGN MODELS				
	- To analyze object-oriented analysis and design models in software development. - To develop software design models using appropriate methodologies, tools, and techniques	Object – Oriented Analysis and Design Models - Procedural Programming - OOP - Development Models - Tools and Techniques	Faculty-Led Discussion: - Explain OOAD concepts, OOP principles, and software development models. Canvas Activities: - Upload UML exercises and participate in discussion forums. Asynchronous Learning through Paraverse: - Watch OOAD M-Flix videos and complete Flashcards. Diagnostic Mapping: - Differentiate procedural programming from object-oriented programming. Adaptive Practice: - Develop UML class diagrams from case scenarios. AI-Assisted Activity: - Generate a UML class diagram using AI, then validate class relationships, multiplicities, and design quality before submitting the revised version.	Mastery Check – OOAD Analysis and Design Mastery Check: (AI-Prohibited) Students individually receive a software system scenario. Technical Assessment 8: UML Modeling Activity (Use Case, Class, or Sequence Diagram). (AI-Assisted) Technical Assessment 9: Object-Oriented Design Report explaining design decisions. (AI-Assisted) Evidence of Learning: The group submits: - UML Modeling - Object-Oriented Design	1, 3
Week 8 / 4 hours	MODULE 6 – DEVELOPMENT STRATEGIES OVERVIEW				
	- To select appropriate software development strategies for different software project scenarios - To construct UML behavioral and structural diagrams	Software Development Strategies - Software Development Strategies Unified Modeling Language (UML) - Behavioral UML - Structural UML	Faculty-Led Discussion: Discuss software development strategies and UML behavioral and structural diagrams. Canvas Activities:	Technical Assessment 10: UML Behavioral and Structural Diagram Portfolio. (AI-Assisted) Technical Assessment 11: Project Planning Exercise (Gantt Chart, PERT Chart, Four Ps) (AI-Assisted)	3

	that satisfy software design requirements		- Submit UML diagrams and participate in peer feedback. Asynchronous Learning through Paraverse: - Complete UML video lessons and Flashcards. Diagnostic Mapping: - Identify the most appropriate development strategy for a project scenario. Adaptive Practice: - Revise UML diagrams based on instructor feedback AI-Assisted Activity: - Use AI to recommend a software development strategy, then evaluate its suitability against project requirements and justify the final selection.	Student Evidence: The portfolio may contain: - UML Behavioral and Structural Diagram Portfolio - Project Planning Exercise	
Week 9 / 4 hours	MODULE 7 – TRENDS IN SOFTWARE ENGINEERING				
	- To Evaluate emerging software engineering technologies based on their applicability to modern software development - To critique current	Trends in Software Engineering - AI-Native Software Engineering - LLM - Green Software Engineering - GenAI - Low-Code - DevSecOps - Cybersecurity - Microservices - Edge Computing	Faculty-Led Discussion: - Discuss AI-Native Software Engineering, LLM-based applications, DevSecOps, Green Software Engineering, Microservices, and emerging technologies. Canvas Activities: - Submit a trend analysis report and participate in an online discussion forum. Asynchronous Learning through Paraverse: - Complete M-Flix lessons on emerging software engineering trends and review Flashcards.	Summative Assessment 3: Emerging Technology Evaluation Report on AI-Native Software Engineering, DevSecOps, or Green Software Engineering. (AI-Prohibited) Technical Assessment 12: Emerging Technology Evaluation Report on AI-Native Software Engineering, DevSecOps, or Green Software Engineering. (AI-Integrated) Technical Assessment 13: Update and Final Thesis Title	3

			Diagnostic Mapping: - Analyze a case involving the adoption of AI-enabled software engineering practices. Adaptive Practice: - Compare AI-generated and human-developed software engineering solutions using a guided evaluation rubric. AI-Integrated Activity: (AI Solution Analysis) - Students are required to use an approved AI tool (e.g., ChatGPT Edu or Custom GPT) to generate a software engineering solution (e.g., system architecture, requirements specification, or development strategy). Students must evaluate the AI-generated solution for correctness, completeness, security, scalability, and alignment with software engineering principles, revise the output, document the prompts used, justify all revisions, and reflect on the strengths and limitations of AI in software engineering practice. - This activity assesses both software engineering competence and responsible AI use, making it appropriately AI-Integrated under the SMART-AIMED framework.	Submission Evidence: - Approved Title Proposal - Ethics Review Form - AI Prompts - AI-Generated Solution/Output - AI Interaction Log - AI Usage Declaration - AI Reflection - Team Contribution Record	
Week 10 / 4 hours	CONSULTATION/ SUBMISSION OF THESIS PROPOSAL				

Week 11-12 / 8 hours	DEFENSE
Week 13-14 / 4 hours	DEFENSE
Week 14 / 4 hours	SUBMISSION OF APPROVED THESIS TITLE

Note:

AI-Prohibited = AI is not permitted for a specified academic activity or assessment

AI-Assisted = AI may be used for specified support functions; students remain responsible for the work and final output

AI-Integrated = AI is intentionally used as part of the learning activity or assessment to achieve the learning outcomes

CLO	EXAMS / ASSESSMENTS
1	Summative Assessment 1, Technical Assessment 1, Technical Assessment 2, Technical Assessment 3, Technical Assessment 6, Technical Assessment 7, Technical Assessment 8, Technical Assessment 9
2	Summative Assessment 2, Technical Assessment 3, Technical Assessment 4 and Technical Assessment 5
3	Summative Assessment 3, Technical Assessment 10, Technical Assessment 11, Technical Assessment 12 and Technical Assessment 13 (<i>Project Title Proposal Presentation and Documentation</i>)

GRADING SYSTEM	
Midterm Grade	Final Grade
Midterm Grade (MG) = 60% Class Standing + 40% Lecture: 100% Midterm Exam Class Standing (CS) 60% - Average of Summative Assessments - Teacher's Evaluation - Class Participation (Seatwork, Assignments, Recitations) - Short Quizzes Midterm Exam (ME) 40% PASSING RAW SCORE: 70	Final Grade (FG) = 60% CS + 15% ME + 25%(Title Proposal) Class Standing (CS) 60% Title Proposal 25% Midterm Exam (ME) 15% Lecture: 100% Class Standing (CS) 60% - Summative Assessments - Project Proposal Midterm Exam (ME) 15% Final Exam (FE) - Defense 25% PASSING RAW SCORE: 70

INSTRUCTIONAL MATERIALS
Course Materials (lecture slides, study guides, supplementary materials, video courseware, etc.) through Canvas Learning Management System (LMS): https://feu.instructure.com Paraverse: https://paraverse.feutech.edu.ph/ Network Map: https://paraverse.feutech.edu.ph/network-map/ M-Flix: https://paraverse.feutech.edu.ph/mflix/

TEXTBOOK/REFERENCES/ONLINE REFERENCES

Online References:

- Sommerville, I. (2020). Engineering Software Products: An Introduction to Modern Software Engineering. ISBN-13 978-0133943030: Pearson Education Inc,
- Chopra, Rajiv. (2021). An Integrated Approach to Software Engineering. ISBN: 978-9350147191
- Tsui, F., Karam, O. and Bernal, B. (2022). Essentials of Software Engineering (5th Edition). ISBN: 9781284228991: Jones & Bartlett Learning
- Farley, D. (2022). Modern Software Engineering: Doing What Works Better to Build Better Software Faster (1st Edition). ISBN: 978-0137314911: Addison-Wesley Professional
- Martin, I. (2024). Software Engineering Step by step: An Introduction to Designing, Developing and Delivering Software. ISBN-13 979-8303309197.

Online Sources:

- Courseware Materials available at <http://feu.instructure.com>
- FEUTECH e-book library collection at <https://drive.google.com/drive/u/0/folders/1Tiu-nHjg8lnnBDU82rsN5qj5WTptz7Bx>

Journals:

MIS Quarterly

Security Management

Tech & Learning

Journal of Educational Multimedia & Hypermedia (AACE)

IEEE

ScienceDirect

ACM

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